Takeo Shimizu and the 'Schools' Cloud Chamber': Research and Pedagogy in the Search to Visualise Atomic Processes, from Cambridge to Tokyo

Modern Sciences

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Introduction

The focal point of this essay is a cloud chamber from the Whipple Museum, made by the Cambridge Instrument Company (CIC) in 1927 based on an earlier design by Takeo Shimizu.[1] Shimizu was a Japanese graduate student at the Cavendish Laboratory, Cambridge, in 1920-1921, modifying the Wilson cloud chamber for research purposes. He patented a design with the CIC before returning to Japan. In 1927, the CIC released a modified version of Shimizu's model for teaching purposes. After introducing some historical context, I will form two arguments regarding the relationships between teaching and research related to cloud chambers, on the basis of this instrument.

Firstly, I will argue this object has been underdiscussed in the literature as a key demonstration of the cloud chamber's relations to pedagogy and research within the 20th Century. Here, I draw upon Bertozzi's 2021 paper to provide an analytical framework, and discuss the ways in which the CIC modified Shimizu's research apparatus for pedagogy. I will argue that Bertozzi's account has overlooked the significance of this particular object as a successful teaching instrument, and a particularly early example of a pedagogical cloud chamber made in Cambridge within an active research period.[2]

Secondly, I will explore how this instrument and Shimizu's career demonstrate how cloud chamber research was enriched both in England and Japan via scholarly exchanges. I will discuss the phenomenon of Japanese graduate students coming to England to study physics before returning to Japan and using their experiences to shape the direction of research in their home country. I will similarly argue that cloud chamber research in England was strengthened through contributions from these graduate students, and further collaborations with Japanese physicists.

[1] Whipple Museum of the History of Science, accession no. 4459.
[2] Bertozzi, 2021, 193.

C.T.R. Wilson and the Early Cloud Chamber

I will now discuss C.T.R. Wilson, the inventor of the cloud chamber and an influential figure at the Cavendish Laboratory, to contextualise cloud chamber research in 1921. While in Cambridge, Wilson realised that within a supersaturated chamber with an absence of dust, water droplets could be made to nucleate immediately about the location of charged particles.[3] Identifying the charge and numbers of these droplets would therefore reveal information about the correlated particles. In 1911, after many years of careful, sustained research, Wilson published the first photographs of atomic tracks using his cloud chamber technique.[4] In his design, the chamber would rapidly expand in tandem with the taking of a flash-bulb photograph.[5] Wilson left the Cavendish in 1919 to take up a position in the Solar Physics Observatory, but kept close ties with the Laboratory, continuing to influence and assist in cloud chamber research there throughout the 1920s.[6]

Takeo Shimizu at the Cavendish

Born in Japan in 1890, Takeo Shimizu graduated from the Department of Experimental Physics at Tokyo Imperial University in 1914.[7] After the war, Shimizu spent two years working with X-rays in William Duane's laboratory at Harvard University, co-publishing five papers with Duane (chair of bio-physics) and presenting some results to the American Physical Society.[8] In 1920, Shimizu took up a position as Ernest Rutherford's first graduate student at the Cavendish Laboratory.[9]

Research at the Cavendish around this period mainly focused on nuclear physics.[10] Upon Shimizu's arrival, Rutherford was concerned with the nuclear disintegration of nitrogen upon bombardment with alpha particles. Alpha particle collisions with nitrogen occur from an

[3] Assmus and Galison, 1989, 244-245, 248-262, 266-269. Barron, 1952, 1-2. Longair, 2016, 122-123. Phillipson, 2016, 35-37. Staley, 2006, 95.
[4] Barron, 1952, 2. Phillipson, 2016, 35-37.
[5] Longair, 2016, 156.
[6] Barron, 1952, 9, 13.
[7] Ito, 2018, 293. For a photograph of Shimizu, see Longair, 2016, 185.
[8] Bridgman, 1936, 25. Duane and Shimizu, 1918, 1919. Stuewer, 2018, 174. The American Physical Society, 1918, 478, 491-492. Hermann Blumglart was influenced by Duane to develop the first radioactive tracers, which he used with a modified Shimizu-type cloud chamber. Patton identifies this as 'the birth of nuclear medicine instrumentation' in Patton, 2003. Exploring this is unfortunately beyond the scope of this essay.
[9] Stuewer, 2018, 174.
[10] Gariboldi and Tucci, 2006, XV.

estimated 0.001% of rays produced by so-called radium C, now known as bismuth-214, a decay product of radium itself.[11] Scintillation methods for particle detection were being commonly employed at the Cavendish, but were unsuccessful here; the likelihood of an alpha collision with nitrogen and subsequent disintegration was too small, and the associated error too large.[12] Indeed, scintillation methods were tiring as they required manual tracking and counting, so often led to mistakes.[13] Therefore, in 1920 Rutherford tasked Shimizu to utilise the cloud chamber to find evidence of these events. Shimizu went on to make two important modifications to Wilson's cloud chamber, summarised within two papers published together in the Proceedings of the Royal Society in 1921: 'A Reciprocating Expansion Apparatus for Detecting Ionising Rays', and 'A Preliminary Note on Branched α-Ray Tracks'.

The first modification concerned the high speed of expansion of Wilson's cloud chamber. This created well-defined tracks, but made it difficult to associate time information with them. Shimizu developed a chamber which instead underwent slow, repeated expansions that could register tracks continuously over hours.[14] This was achieved by mechanically connecting the piston of the cloud chamber, responsible for its expansion, to a 'prime mover' (i.e. a handle which was turned manually) instead of moving it via air pressure.[15] Shimizu was able to achieve rates of 50-200 expansions per minute.[16] In Rutherford's 1920 Bakerian Lecture to the Royal Society, he noted this enabled himself and Shimizu to visually identify branched tracks: a promising sign.[17]

Due to the rarity of these events, it was unrealistic sufficient evidence could be found by direct observation. Instead, a large number of photographs would have to be taken.[18] As Rutherford phrased it in his Bakerian Lecture, '[e]ye observations of this kind are too uncertain to regard them with much confidence, so arrangements are being made by Mr.

[11] Blackett, 1959, 294. Chaloner, 2016, 361. Longair, 2016, 200. Longair, 2021, 449. Patton, 2003, 1363. Rowlands and Todd, 2021, 97.
[12] Barron, 1952, 8. Blackett, 1948, 293-294. Blackett, 1959, 97. Campbell, 2021, 63. Kim, 2002, 160. Rowlands and Todd, 2021, 97. Shimizu, 1921a, 432, 435.
[13] Chaloner, 1997, 358, 361, 364, 368, 372. Longair, 2016, 179, 197-199. For a more detailed account, see Hughes, 1993.
[14] Barron, 1952, 8. 'Cambridge Ray-Track Apparatus', 1927, 2. Leone and Robotti, 2008, 177-178. Shimizu, 1921b, 426. Shimizu, 1922, 1.
[15] Barron, 1952, 8. Blackett, 1959, 294. Kim, 2002, 160. Rowlands and Todd, 2021, 97-98. Shimizu, 1921b, 426.
[16] Barron, 1952, 9. Cattermole and Wolf, 1987, 78, 255. Shimizu, 1921b, 426.
[17] Rutherford, 1920, 393.
[18] Blackett, 1959, 294. Rowlands and Todd, 2021, 97. Stuewer, 2018, 175. Only eight of Blackett's twenty-three thousand photographs demonstrated this effect.

Shimizu to obtain photographs, so that the tracks can be examined in detail at leisure'.[19] As I will return to later, this demonstrates attitudes towards photography as a research necessity at the Cavendish in this period. Shimizu thus incorporated an automatic camera to take numerous photographs of alpha ray tracks.[20] The repeated expansions also allowed more photographs to be taken within a given time.[21] Therefore, Shimizu both created this new form of cloud chamber with repeated expansions, and improved its photographic component. Phillipson has identified this as an early step towards modern particle physics, with the creation and subsequent analysis of large amounts of data.[22] This is a statement as to the importance of the work Shimizu was doing.

In 1921, Shimizu suddenly returned to Japan due to family problems.[23] Rutherford assigned another graduate student, Patrick Blackett, to continue Shimizu's research.[24] Shimizu's two papers were communicated on June 4th of that year, and Blackett's laboratory notebooks suggest Blackett began his work in July.[25] By 1924, Blackett had increased the number of photographs that could be taken in the chamber, improved the quality of the images, fully automated the reciprocation process with a spring mechanism, and found traces of alpha particles undergoing nuclear collisions.[26] In 1925, Blackett published that with cloud chamber photographs, he had succeeded in finding visual evidence of the disintegration of nitrogen upon bombardment with alpha particles.[27] Blackett also, alongside Giuseppe Occhialini, went on to make significant modifications which allowed for detection of cosmic rays.[28] The 1948 Nobel Prize in physics was awarded to Blackett for both of his discoveries made using cloud chambers.[29] This shows the importance of the research track that Shimizu was pursuing. Indeed, Blackett acknowledged Shimizu's contribution in his Nobel Speech.[30]

[19] Rutherford, 1920, 393. In Chaloner, 1997, 361.
[20] Phillipson, 2016, 37. Shimizu, 1921b, 425.
[21] Shimizu, 1921a, 432.
[22] Phillipson, 2016, 37. See also Chaloner, 1997, 365, and Assmus and Galison, 1989, 226.
[23] Blackett, 1948, 97. Blackett, 1959, 98. Campbell, 2021, 63. Stuewer, 2018, 175.
[24] Blackett, 1948, 97-98. Blackett, 1960, 293. Bustamante, 2006, 39. Campbell, 2021, 63. Gariboldi and Tucci, 2006, XV. Kim, 2002, 160. Longair, 2016, 200. Stuewer, 2018, 175. Yagi, 1973, 347.
[25] PB/2/1/2, The Royal Society Archives. Shimizu, 1921.
[26] Blackett, 1948, 98. Blackett, 1960, 293. Bustamante, 2006, 39. Longair, 2016, 200-201.
[27] Bertozzi, 2021, 187. Blackett, 1925. Blackett, 1960, 293. Longair, 2016, 200-201.
[28] Barron, 1952, 12. Bertozzi, 2021, 184. Leone and Robotti, 2008, 180-184.
[29] Bertozzi, 2021, 188.
[30] Blackett, 1948, 97-98.

The Cambridge Instrument Company

In the forty years since its initial founding as the Cambridge Scientific Instrument Company, the CIC became a significant supplier of apparatus for the Cavendish.[31] Shimizu patented his designs with the CIC for commercial manufacture in February 1921 under a royalty agreement.[32] This was fairly usual practice. Teaching staff such as Wilson worked with the CIC to manufacture their designs, and Wilson was an important contact to help fulfil the commercial and educational roles of apparatus developed at the Cavendish.[33] Given Wilson's influential role, it is understandable Shimizu patented his design with the same company.

The CIC continued manufacturing Shimizu's 1921 'Ray-track Apparatus', described in the trade literature as follows: 'Mr. Takeo Shimizu, of Japan, working at the Cavendish Laboratory, Cambridge, has recently introduced some important modifications in the design of Mr. Wilson's original instrument… by means of which the continuous emission of *alpha* and *beta* particles can be conveniently watched… observed and photographed'.[34] In one 1926 advertisement in the *Philosophical Magazine*, an image of the apparatus accompanies the following caption: '[t]his instrument embodies Mr. T. Shimizu's modification of Prof. C.T.R. Wilson's cloud expansion apparatus. It enables the continuous emission of ionising particles to be observed and photographed'.[35] However, in this same year, the company began to design a new apparatus based upon the Shimizu model, but explicitly for educational needs. In 1927, it was released as the 'Schools' Cloud Chamber', or the 'Cambridge Ray-Track Apparatus'.[36] The addition of the word 'Cambridge' may reflect the company's desire to assert their role in its design.

As the CIC was unable to locate Shimizu in Japan, it seems he never received the royalties due to him.[37] This is somewhat strange, given the correspondence between physicists in Cambridge and Tokyo I later discuss. While they certainly credited Shimizu within the

[31] Longair, 2016, 49, 93.
[32] Cattermole and Wolf, 1978, 78, 255. Kim, 2002, 156. Longair, 2016, 200. Shimizu, 1922.
[33] Kim, 2002, 153, 155-156.
[34] 'Ray-track Apparatus', 1922, 2.
[35] Cambridge Instrument Company advertisement (1926) *Philosophical Magazine*, 1. In Chaloner, 1997, 366-367.
[36] Barron, 1952, 10-11. 'Cambridge Ray-Track Apparatus', 1927, 2. Cattermole and Wolf, 1978, 257. Chaloner, 1997, 367.
[37] Cattermole and Wolf, 1978, 78.

literature surrounding the 1921 and 1927 models, the implied lack of effort in tracking him down seems potentially exploitative.

This brings me to the object serving as the focal point for this essay. While not directly designed by Shimizu, it operates on the same basis of continual reciprocating expansions. Turning a handle turns a clockwise wheel to move a piston, which causes the chamber to expand and the air in the chamber to become supersaturated. A radioactive source on a pin provides alpha particles which move through the chamber and create tracks. These are viewed through a circular glass plate, using replaceable filter papers blackened with Indian ink and illuminated by an adjacent filament bulb.[38] As the piston is reciprocating, tracks form continually as the handle is turned. An electric field clears the chamber of ions between subsequent expansions.

This object represents a direct attempt by the CIC to transform a research instrument into a pedagogical one. The company's attitudes towards pedagogical instruments at this time is summarised in the following quotation from a 1927 catalogue containing this model: 'measurement and demonstration of physical phenomena is not only of interest to the scientist and the research worker, but occupies an important place in the curriculum of modern schools and colleges… to obtain the maximum educative value from the work carried out in school laboratories, it is important to ensure that they are adequately equipped with apparatus designed upon scientific principles and of sufficiently robust construction to give accurate results in the hands of the average student.'[39] I will thus consider why Shimizu's design was chosen, what modifications the CIC made, and what this tells us about the role of the cloud chamber in pedagogy. As Cattermole and Wolfe phrase it, '[c]omparison with Wilson's experimental apparatus provides an interesting illustration of the way a piece of laboratory apparatus may be refined into a relatively simple piece of equipment to demonstrate scientific phenomena in a manner which can be understood by those who have only a limited knowledge of science'.[40]

Firstly, I will address why Shimizu's research design was utilised for the CIC's pedagogical purposes. Shimizu's reciprocating expansion design was inferior to Wilson's (with its single,

[38] 'Cambridge Ray Track Apparatus', 1927, 3.
[39] 'Cambridge Physical Instruments', 1927, 22.
[40] Cattermole and Wolfe, 1978, 257.

sudden expansion) in terms of the clarity of the particle tracks, as previously mentioned. However, it allowed for continual production of tracks, and thus a continually demonstratable educational model. The importance of track clarity is also lessened outside of research, given the purpose of an educational demonstration that such particles exist only requires tracks to be produced.

The CIC simplified Shimizu's design in various ways. The chamber has a fixed expansion ratio, meaning initial and final volume of the chamber for an expansion are the same.[41] It also only produces tracks for alpha particles, and not beta particles. Proof of a single type of charged particle (although not a fundamental one) seems understandably sufficient for an educational device of the period. The chamber is also smaller than Shimizu's own patented research model, making it cheaper to produce.[42] It is thus a more pragmatic choice for a pedagogical model: more affordable, and easier to transport and store within smaller institutions.

One way the chamber was made smaller was by removing the camera function.[43] Considering the explicit mention of photography in the 1921 trade literature and 1926 advert previously discussed, this seems to have been a conscious decision for a teaching model. The Rutherford quotation from his Bakerian lecture exemplifies how photographs took on a significant research function within this period, presenting tangible, verifiable evidence of particle tracks.[44] Cloud chamber research in the early 20th Century closely tailed photographic developments.[45] Part of Shimizu's contribution to English research was based on photography and the use of the camera within cloud chambers. Indeed, this is how Blackett summarises Shimizu's work in his 1925 paper on the disintegration of nitrogen: '[t]he camera, designed originally by Shimizu, takes two photographs at right angles on standard cinematograph film'.[46] For educational purposes, however, the photography aspect was relatively unimportant. Visual evidence alone was sufficient.

[41] Barron, 1952, 10-11. Cattermole and Wolf, 1978, 257. Supersaturation and nucleation about charged particles will only occur at and above a certain expansion ratio in a Wilson cloud chamber.
[42] Barron, 1952, 10-11.
[43] 'Cambridge Ray-Track Apparatus', 1927, 2.
[44] Rutherford, 1920, 393. In Chaloner, 1997, 361.
[45] Chaloner, 1997, 368.
[46] Blackett, 1925, 350.

Chaloner offers a different perspective, arguing the pedagogical significance of photographs which represent 'mechanical witnesses' to atomic events.[47] He even suggests that the production of such photographs rendered the purchase of a cloud chamber for educational purposes somewhat unnecessary. How, then, do we understand the decision of the CIC to produce a schools model? Firstly, they seem to have been responding to actual demand. As phrased in the sales literature, '[i]t is found, however, that among schools, colleges, and technical institutes there is a demand for a simpler form of apparatus which renders the phenomena easily visible, but which does not necessarily enable photographs to be taken of the tracks'.[48] Secondly, and to understand this demand, I argue that the process of actively participating within the experimental process, and the physicality of using the instrument, serves an important pedagogical function for inspiring and engaging students. The addition of the handle both makes the apparatus easy to use, and physically involves the user within the experimental process. This contrasts with the relatively detached process of looking at a photograph. Chaloner himself acknowledges that '[t]he camera apparently eliminated human agency'; I wish, then, to draw attention to the importance of human agency in education, as direct interaction with experiment creates inspiration and interest.[49]

The success of the 1927 model as a pedagogical instrument can be seen from its longevity. It sold relatively well, and was still in production in 1951 for the Festival of Britain exhibition, thirty years after Shimizu left the Cavendish.[50] An estimated 3000 people per day over a five month period used this model.[51] In fact, the CIC developed another pedagogical model on the basis of Shimizu's designs for the exhibition, able to run continually due to multiple, recurrent expansions of the chamber, yet instead driven by a motor.[52] This latter model seems to be particularly suited for the exhibition, as visitors could walk by and see the continual production of tracks – not very useful for research, but engaging for a pedagogical display. Interestingly, Chaloner describes this as the 'apotheosis of the cloud chamber as a pedagogic device', whereas the 1927 model as it was in the 1920s is only afforded a brief mention. Perhaps this is because of the sheer number of people who interacted with the apparatus.

[47] Chaloner, 1997, 367.
[48] 'Cambridge Ray-Track Apparatus', 1927, 2. 'Cambridge Physical Instruments', 1927, 22.
[49] Chaloner, 1997, 368.
[50] Barron, 1952, 11. Cattermole and Wolf, 1978, 78, 257. Phillipson, 2016, 37.
[51] Cattermole and Wolf, 1987, 79. Phillipson, 2016, 37.
[52] Barron, 1952, 13. Cattermole and Wolf, 1987, 79.

However, I would like to more closely examine the significance of this object in the active research decade and location in which it was created.

Bertozzi and the Pedagogical History of the Cloud Chamber

Bertozzi, in his 2021 article, draws attention to the secondary life scientific instruments often find within teaching after existing as cutting edge research apparatus, a phenomenon more commonly discussed regarding 19th Century objects.[53] He similarly argues creating broad, historiographical distinctions between research and pedagogy can be 'unsustainable'.[54] Bertozzi thus draws attention to three case studies, but not this instrument, which I argue overlooks a significant development within the pedagogical history of the cloud chamber.

The CIC's appropriation of Shimizu's design in the 1920s is unusual by Bertozzi's account, as he identifies the 1930s as the significant period for the popularisation of cloud chamber teaching models.[55] The 'Schools' Cloud Chamber' seems particularly comparable to Bertozzi's discussion of Carlo Ballario's efforts in the late 1950s-early 60s to create a model which was easy to use and understand, yet precise in the same manner as research instruments.[56] Ballario created an apparatus with three key differences to the CIC's model. It continually produced tracks of high precision using a pressure-driven mechanism, rather than manual expansions of the chamber. This was possible from 1933 due to Wilson's research efforts (so unavailable to the CIC in 1927). Ballario's model relied heavily on electronics, operated by a button and dial, so did not utilise the physicality of the handle. It also had a camera function. Ballario's instrument had a much shorter lifespan than the 'Schools' Cloud Chamber', only distributed between 1959 and 1962. This, Bertozzi states, is probably because it was delicate and hard to use. Bertozzi seems particularly interested in this model because of Ballario's intentions to introduce 'the daily effort that experimental physics endures every day to provide evidences to the great theoretical frameworks' into a teaching environment. Given this was Ballario's intent, it was perhaps misguided to focus on track precision rather than the manual operation of the instrument.

[53] Bertozzi, 2021, 179.
[54] Bertozzi, 2021, 193.
[55] See also Phillipson, 2016, 37.
[56] Bertozzi, 2021, 180, 188-189.

The museum object represents a very early attempt to create a pedagogical cloud chamber, and (at least in some respects) a more successful one than Ballario's. Bertozzi situates the 'golden age' of cloud chamber research in the period between the publication of Wilson's images of the cloud chamber and the first cloud chamber atlas in 1940.[57] Therefore, the 1927 pedagogical model was created in a time and place more commonly noted for research. Blackett emphasised the value of the cloud chamber as a teaching instrument which allowed atomic processes to be 'easily visualised and understood without the aid of complicated mathematics or mastery of deep theories'.[58] This is suggestive of attitudes towards the significance of the cloud chamber in pedagogy as well as research at the Cavendish. In light of this, I argue that a pedagogical history of the cloud chamber benefits from inclusion of this object, which directly exemplifies significant interrelationships between research and pedagogy.

Shimizu in England and Japan

Shimizu travelling to Cambridge can be understood both within Japanese tradition, and as part of the culture at the Cavendish during the 20th Century to receive visiting scholars. I will now argue that scholarly exchange created mutual strengthening of cloud chamber research in England and Japan.

The tradition for Japanese scholars to study abroad began within the Meiji period, and allowed for enrichment of research culture in Japan.[59] In 1880-1920, 54 Physics doctorates were awarded in Japan.[60] Despite the relative rarity of physics scholars around this time, Shimizu was not alone in coming to England as a graduate student. Noted physicist Yoshio Nishina (co-author of the Klein-Nishina formula, which describes photon-electron scattering) was born in the same year as Shimizu, and similarly came to the Cavendish for around a year, arriving in the autumn of 1921 and performing experiments on Compton scattering.[61] Professor Hantarō Nagaoka of Tokyo Imperial University arranged this with Rutherford.

[57] Bertozzi, 2021, 179, 183.
[58] Blackett, 1952, vii. In Bertozzi, 2021, 188.
[59] Bartholomew, 1989, 68, 75-76, 92. Brown and Rechenberg, 1996, 97. Morris-Suzuki, 1994, 66, 82. Schopper, 1996, 5.
[60] Bartholomew, 1989, 51-52.
[61] Kubo, 2008, 18.

Perhaps he also organised Shimizu's year abroad, as Shimizu had been one of his pupils.[62] Similarly, Suekichi Kinoshita discovered a method to visualise atomic processes using alpha particles and photographic film while Rutherford's graduate student at the University of Manchester in 1909. He later returned to Japan and continued his career at Tokyo Imperial University using radium donated by Rutherford, looking at photographic film processes and, by the 1920s, cloud chambers.[63] He also developed the first university course in radioactivity within his home country.[64] Three related papers from Kinoshita can be found in Wilson's archives, a further testament to the teaching and research links between England and Japan within this field.[65]

Back in Japan, Shimizu continued his work with nuclear research and radioactivity.[66] In 1922, he published a paper on modifications to an electroscope designed by Wilson.[67] The influence on Shimizu's career here is obvious. In 1926, Shimizu became chair of Theoretical Physics at Tokyo Imperial University, meaning his role shifted from research student to lecturer relatively quickly.[68] For comparison, it was 1933 when Blackett became a professor at Birkbeck College in London. It is notable Shimizu was chair of theoretical, not experimental physics, even though Shimizu was trained in the latter. At this time, theoretical physics was somewhat an emerging discipline, with less prestige than its experimental counterpart, a potential reason I have not found much research from this period.

Other papers that Shimizu published in Japan include 'An Elementary Deduction of Einstein's Law of Gravitation' in 1924.[69] In 1926, he wrote a paper on mercury in glass capillaries, publishing it in both English and Japanese.[70] This variety, in topic and in language, seems to demonstrate a keenness to participate at the frontiers of physics research. In 1928, he gave a lecture on the relevance of electron theory and quantum mechanics to

[62] Ibid. Shimizu, 1925. The Nagaoka Anniversary Volume was donated to the University of Cambridge by Tokyo Imperial University, again demonstrating scholarly exchanges between them.
[63] Bartholomew, 1989, 178. Chaloner, 1997, 373-374. Ikeuti and Kinoshita, 1915. Ikeuti, 1916. Ikeuti, Kinoshita and Akiyama, 1921. Yagi, 1973, 347.
[64] Tanaka and Yamasaki, 1986, 141.
[65] CTRW4/659-661, Churchill Archives Centre, University of Cambridge.
[66] Ito, 2018, 308. Low, 2005, 146.
[67] Shimizu, 1922.
[68] Ito, 2018, 308.
[69] Shimizu, 1924.
[70] Shimizu, 1926.

electrical engineering.[71] This was published in the Journal of the Institute of Electrical Engineers of Japan as 'On the History and Recent Advancement of Electron Theory'.[72] The title alone suggests aims to spread cutting edge developments in physics. His choice of journal is also notable, as he is specifically addressing electrical engineers in Japanese, suggesting an interest in shaping the direction of Japan's industry. Similarly, he was on the subcommittee for the Kyoto International Conference on Electronics in 1939.[73]

Correspondence to Wilson from physicist Ukichiro Nakaya in 1935 provides another insight into Shimizu's life.[74] Nakaya was working at Hokkaido Imperial University in meteorological physics. Given Wilson's research interests, it is perhaps unsurprising to find they were in communication. Wilson was assisting Nakaya in publishing a paper in the Proceedings of the Royal Society. In one letter, Nakaya's phrasing suggests he is responding to Wilson's enquiries after Shimizu. He informs Wilson that unfortunately, Shimizu had been very sick with what they had thought was tongue cancer, but had been found to be non-cancerous. Nakaya informed Wilson that Shimizu intended to resign from Tokyo Imperial University, as he could no longer give lectures. This may be why it is difficult to find information on Shimizu in this period and later.

This correspondence is also indicative of relevant links between England and Japan. While gaining scientific knowledge from Nakaya, Wilson was able to help give Nakaya's journal article a more widespread audience. It also seems significant that Wilson was actively enquiring after Shimizu 14 years after his return to Japan. Shimizu was not a direct student of Wilson's, yet his research was influenced by him, and this enquiry is suggestive of their positive relationship.

In the 1930s, Shimizu led a cloud chamber research project at the Institute of Physical and Chemical Research (RIKEN), with Ryōkishi Sagane as one of his students.[75] Sagane in his own career went from experimental physicist to a significant player within Japanese nuclear policy.[76] Sagane was a graduate student at Tokyo Imperial University from 1929 to 1931,

[71] Shimizu, 1928. In Ito, 2018, 308.
[72] Ito, 2018, 308.
[73] Yagi, 1973, 595.
[74] CTRW3/2.
[75] Hayakawa, 1983, 86-87. Low, 2005, 159.
[76] Low, 2005, 158-160.

studying directly under Shimizu. In the 1930s, Sagane also spent time at the Cavendish.[77] It is easy to postulate Shimizu's influence on Sagane, both on the cloud chamber research that Sagane undertook, and Sagane's decision to travel to Cambridge. Sagane's team reported that they had been successful on the 17th of November 1933, mere months after the development of such an apparatus by Blackett and Occhialini.[78] This suggests how closely the research teams in both countries were competing, and Shimizu's pedagogical role in shaping Japanese cloud chamber research.

In 1946, Shimizu oversaw the establishment of the Physical Society of Japan (JPS).[79] JPS arose as one splinter group of the Physico-Mathematical Society of Japan, founded in 1877 under a different name, of which Shimizu was the last President.[80] It is interesting to consider Shimizu's motivations. He may have been part of a movement to further Japanese technological development after the war, emphasising more deeply original technologies over imported ones.[81] His experiences with the American Physical Society and Royal Society may have been influential. An index by Nichigai Associates states that the Society's associated journal was 'established… for the express purpose of reporting Japanese research to the international academic world'.[82] Given Shimizu's role, it seems reasonable to assume he shared these aims.

The phenomenon of visiting graduate students coming from Japan also provided benefits to cloud chamber research in England. I have already discussed the importance of Shimizu's contributions both directly to research and indirectly to pedagogy, as the museum object demonstrates. The labour of Japanese graduate students like Nishina, Sagane and Kinoshita was similarly valuable in terms of discovery and development of ideas. Kim describes the Cavendish of the early 20th Century as the 'true mecca of experimental physics', both attracting and relying upon international talents.[83] In 1895, a change in rules meant that students with undergraduate degrees from other universities could come to Cambridge as graduate students.[84] Kim writes how significant these non-Cambridge graduates were within

[77] Ibid.
[78] Hayakawa, 87.
[79] Schopper, 1996, 5.
[80] Ibid.
[81] Bartholomew, 1989, 1. Morris-Suzuki, 1994, 157.
[82] Nichigai Asoshiētsu, 1975, i.
[83] Kim, 2002, 160.
[84] Longair, 2016, 103, 126-127. Longair, 2021, 445.

the research community at the Cavendish.[85] They would be almost guaranteed to be excellent students in order to win funding, and the funding they brought also contributed to general research successes.[86] By discussing Shimizu, I have aimed to situate this phenomenon within cloud chamber research in England and Japan.

Conclusion

Within this essay, I have discussed both the 'Schools' Cloud Chamber' released by the CIC in 1927, and Takeo Shimizu, whose designs this apparatus was based upon. My first aim has been to argue that the intimate relationship between pedagogy and research is exemplified by examination of this instrument at the Cavendish in this early period. This has been underacknowledged thus far in the historical literature. As well as this, by discussing Shimizu's life and subsequent career in Japan, as well as Nishina, Kinoshita, Nakaya, Blackett and Sagane, I hope to have demonstrated the mutually beneficial international collaboration between these two countries with regards to cloud chamber research.

[85] Kim, 2002, 161. See Longair, 2016, 34, 39, 54, 103.
[86] Longair, 2016, 126-127.

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